

Ethan Batiste

Design Portfolio: <https://ethanwillbatiste.github.io/>

ethan.will.batiste@gmail.com

Education

University of Toronto, Canada

Sept 2025 - May 2030

BAsc in Engineering Science, GPA: 3.73/4.00

Recipient of University of Toronto Scholars Award

Technical Skills

CAD & Prototyping: SolidWorks, Onshape, 3D Printing, Microfluidics, Arduino, Raspberry Pi, 6-DoF Robotics

Programming & Data: Python, C, MATLAB, Git, FEA, NumPy, Pandas, Algorithmic Latency Optimization

Wet Lab & Bio-Tools: PyMol, PLIP, SELEX, PCR, NGS, Multi-Channel Fluorescence Microscopy, BiofilmQ, ImageJ

Professional Experience

Visiting Researcher | University of Waterloo Medical Microrobotics Lab Apr - Sept 2026

- Engineered bio-hybrid microrobots for localized drug delivery, utilizing bovine spermatozoa
- Designed microfluidic attachments and mounts for a 6-DoF robot.
- Quantified *E. coli* biofilm disruption using advanced Z-stack fluorescence time-series imaging.
- Managed cryogenic cell handling and presented novel experimental proposals to the research group.
- Established SOMMeRS 2026, leading a UWaterloo outreach program for high school student lab placements

Junior Scientist | MOLwise BioWorks Inc.

Jun - Aug 2026

- Executed high-throughput assays (NGS, PCR, SELEX) and extracted molecular interaction profiles via PyMol/PLIP to refine AI model accuracy.

Visiting Researcher | U. Waterloo Medical Microrobotics Lab

Jun 2024 - Apr 2025

- Engineered nanoparticle-conjugated sperm bio-hybrids for localized drug delivery, presenting findings across national academic forums.
- Fabricated gelatin nanoparticles using literature-adapted protocols while coordinating cross-functional lab communications.

Projects & Leadership

Real-Time Microhematuria Detection Farcas Lab - UTBIOME

Sept 2025 - Apr 2026

- Built a Raspberry Pi RTSP image server (C/Python) for real-time blood cell detection, optimizing latency via Big-O analysis.

Student-Led Innovations in Medical Microrobotics (UWaterloo)

Sept 2026 - Aug 2026

- Presented novel surgical microrobotic applications for abdominal adhesion relief to aspiring researchers.

University of Toronto Humanoid Robotics | Mechanical Team Member

Sept 2026 - Present

- Currently working with a team to design the hands and feet of a humanoid robot using the iterative engineering design process and course concepts from my education.

Presentations and Publications

SOMMeRS 2026 Poster Presentation - Sperm-Powered Antibiotic Treatment of E. coli Biofilm[Click](#) to view the poster.

WCET 2026 Abstract Coauthor - Wireless microrobotic phosphate delivery for noninvasive calcium oxalate stone dissolution

References available upon request

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